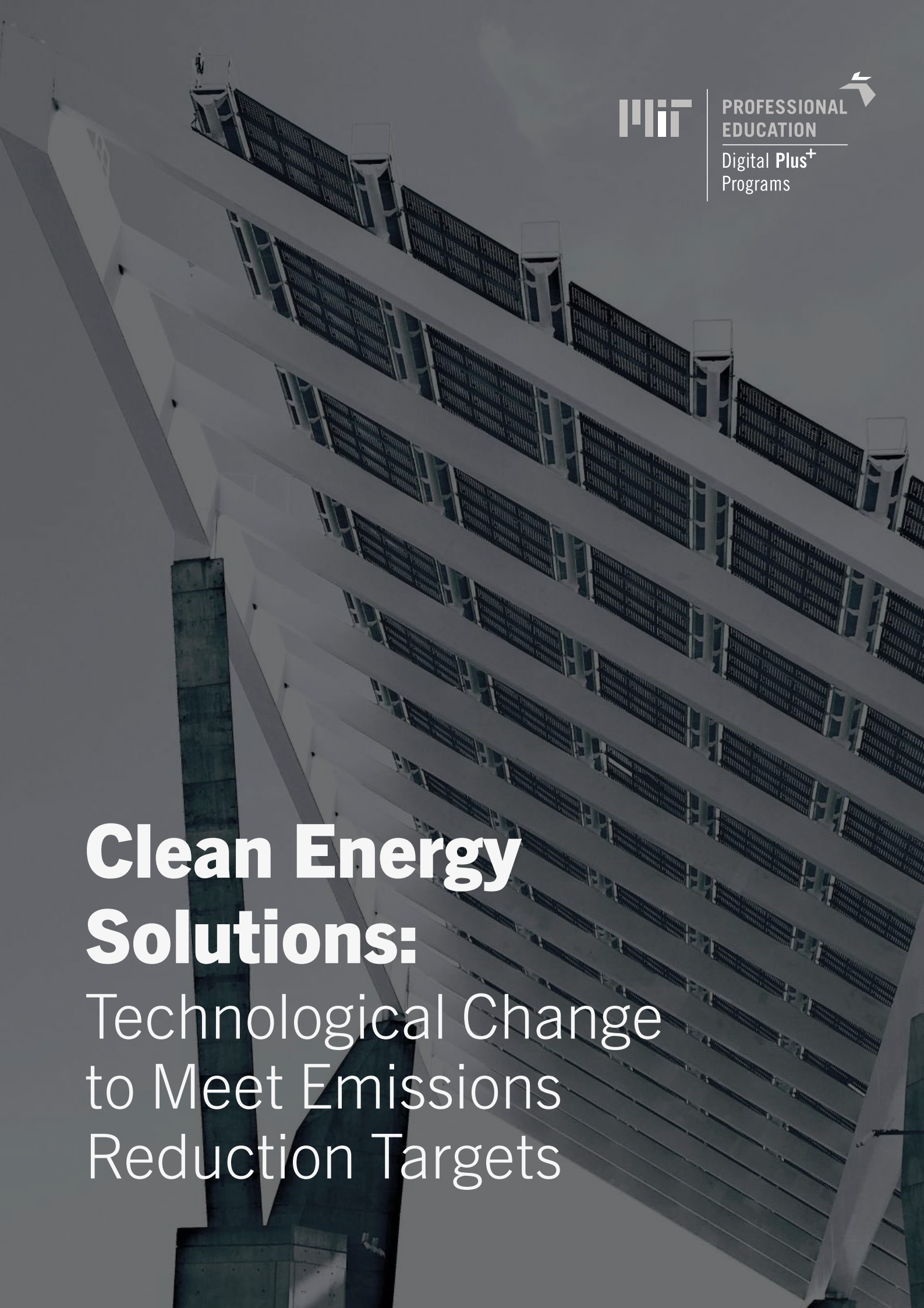
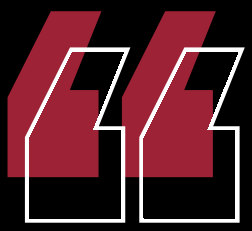


PROFESSIONAL EDUCATION

Digital Plus+ Programs

Clean Energy Solutions:

Technological Change to Meet Emissions Reduction Targets



Mitigating climate change is unavoidably linked to developing affordable, low-carbon energy technologies that can be adopted around the world.

Prof. Jessika Trancik

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Why focus on data-informed clean energy innovation?



Despite increasing awareness of climate change, greenhouse gas emissions have continued to rise relentlessly. But to avoid the worst effects of climate change, emissions need to fall faster than they have risen for over a century.

Nations and an increasing number of organizations around the world have set emissions reduction goals to spur progress toward meeting climate change mitigation targets. However, questions remain regarding the best approach to reducing emissions in a rapid, cost-effective way, while also promoting innovation that benefits economies around the world.

Finding effective strategies for climate change mitigation through data-informed analysis is essential.
Using these insights to make strategic decisions will help secure a healthy future for our planet by supporting a transition toward clean energy systems.

If done right, this transition will offer opportunities for economic growth, job creation, and a more sustainable future for us all.



Being deliberate about clean energy technology—both hardware and software—will play an essential role in addressing climate change and creating economic opportunities that help support human well-being around the world.

About this Course



MIT Professional Education's online course, Clean Energy Solutions, invites professionals to analyze:

1. **Energy systems and technologies**
2. **The options for decarbonization**

Participants will use data to evaluate these systems, model emissions reduction frameworks within their own organizations, and strategize about the development and adoption of clean energy technologies.

10 weeks | 8-10 hours a week | Online

Key Takeaways

This course presents a holistic and quantitative picture of climate change solutions—including hardware —to meet emissions reduction targets.

As participants leverage data to analyze and compare energy systems against climate mitigation goals, they will also strategize about key innovation opportunities for their organizations. Throughout this course, you will:

01 Analyze the scale of the climate change mitigation challenge using current and historical contexts of energy supply and demand technologies.

02 Compare energy conversion and storage technologies based on performance metrics such as cost, carbon intensity, and resource limitations.

03 Analyze data on technological change trajectories.

04 Assess a variety of approaches currently taken to model carbon emissions reduction scenarios and the role of energy systems.

05 Analyze technological change and optimize technology portfolios for your organization.

06 Estimate the rates of change needed in determinants of greenhouse gas emissions.

07 Identify effective strategies for innovation on clean energy and other climate solutions.



Participant Profile

This course is designed for professionals from a multitude of backgrounds who are interested in climate change mitigation efforts and innovation opportunities.

In this course, participants will examine the numbers and science behind climate change mitigation targets and will analyze engineering, policy, and business solutions. Professionals from various roles will benefit from the expertise and skills shared through this course, including:

Engineers and technical professionals looking to take a data-informed approach to devising climate change mitigation solutions that are informed by an understanding of the role and evolution of various energy technologies.

Policy changemakers and advisors at local, national, and international levels who are interested in acquiring data-driven insights into the climate change mitigation opportunities found in clean energy technologies and beyond.

R&D, finance, and investment managers seeking insights into how to best optimize their energy-related portfolios through better anticipation of technological breakthroughs in the field.

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Instructor



Prof. Jessika Trancik

Professor, Institute for Data, Systems, and Society, MIT

Jessika Trancik is an associate professor in the Institute for Data, Systems, and Society at the Massachusetts Institute of Technology. Her research examines the impacts of technologies and the reasons behind technological change.

She has developed theory and predictive models to understand why some technologies improve faster than others, and what technology features enable rapid innovation.

Trancik has developed models for forecasting technological change, which inform engineering design, public policy, and investment portfolios. Several of her theories and models have been applied to new and developing energy technologies, such as solar energy and batteries, and to electricity and transportation systems. Her models have also been used to inform government innovation policy, and applied in diverse industries, including finance, healthcare, manufacturing, software, and consumer products. Her work has been published in journals such as Nature, Proceedings of the National Academy of Sciences, Nature Energy, Nature Climate Change, and Environmental Science and Technology, and has been featured by news outlets such as the New York Times, Washington Post, Financial Times, and NPR.Washington Post, Financial Times, and NPR.

> “To succeed in reducing global greenhouse gas emissions, one should use the most powerful tool for amplifying the benefits of each investment: technology innovation.“

Professor Trancik received her B.S. from Cornell University and her Ph.D. from the University of Oxford as a Rhodes Scholar.

Trancik develops data-informed models to evaluate the economic and environmental impacts of energy technologies over time and space.

Trancik’s models for forecasting technological change have informed engineering design, public policy, and investment portfolios.

Course Outline

Module 1
Introduction

1. Course overview
2. Introducing climate change: How greenhouse gas emissions cause climate change and its impacts society and the environment
3. Climate change mitigation goals: Recent progress and how far we have to go
4. Climate change mitigation goals and recent progress

Module 2
Emissions scenarios and climate Change

1. Background on climate change: The carbon cycle, the greenhouse effect, emissions trends, and the impacts of climate change
2. Anthropogenic emissions sources: A review of greenhouse gas emissions by region and economic activity
3. Looking forward: Establishing a global carbon budget to guide climate policy and international cooperation

Module 3
Evaluation of energy systems

1. Forms of energy and energy technologies: A review of primary and secondary energy sources, energy conversion and storage technologies, and energy services
2. Performance metrics: How measuring the performance intensity and scalability of energy technologies can inform mitigation strategies
3. Performance targets: Three studies analyzing the decarbonization potential of renewable energy technologies, electric vehicle batteries, and electric vehicle charging infrastructure
4. Taking stock of available clean energy solutions: Which energy services need the most innovation to be decarbonized?

Module 4
Technology change trajectories in energy

1. Overview of technological change and data-informed models: Assessing the rapid progress of photovoltaic solar panels, wind turbines, lithium-ion batteries and other energy technologies
2. Data-driven models for measurement and forecasting: Applying Moore's Law, Wright's Law, and other models to forecast future progress of energy technologies
3. Mechanistic models: How the characteristics of a technology enable or inhibit innovation

Module 5
Models to inform investments in clean energy and other climate solutions

1. Technology portfolio models: Applying insights from Modern Portfolio Theory to the challenge of developing clean energy solutions
2. Integrated assessment models: Combining models of climate change with global human and natural systems

Module 6
Energy and climate change policy

1. Overview of energy and climate policy instruments: Which policies are available, which have been adopted, and what are the costs?
2. Regulation: Government mandates for cleaner economic activities
3. Subsidies and investments: Promoting decarbonization with public funds
4. Market-based approaches: Incentivizing economic actors to cut emissions
5. Applying course concepts to policy design: Approaching climate policy as a technology portfolio problem

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Certificate of Completion

All participants who successfully complete the course will receive an MIT Professional Education Certificate of Completion. In addition, they will also earn 8 Continuing Education Units (MIT CEUs). *

In order to obtain CEUs, participants must complete a required CEU confirmation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

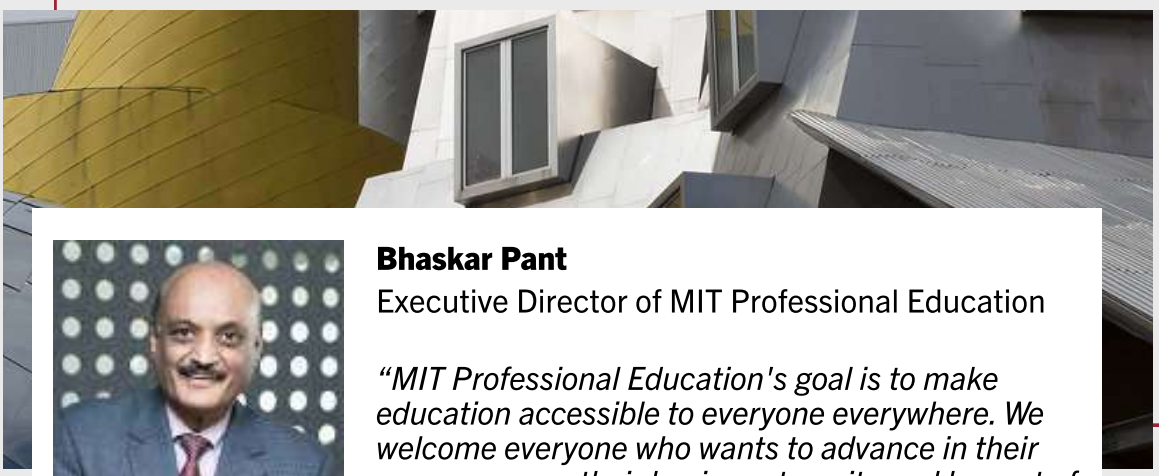
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

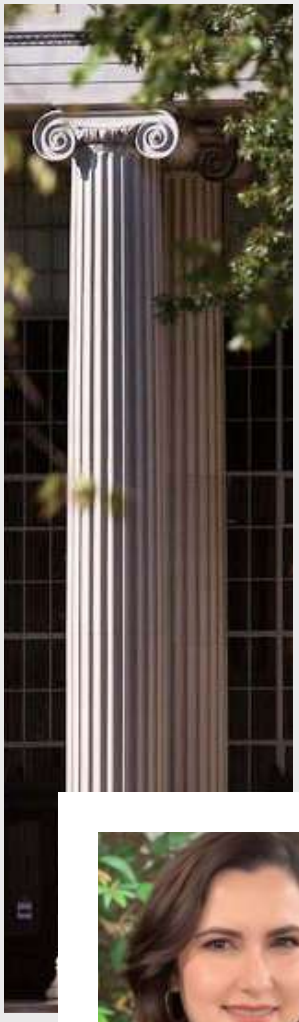


For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience.**

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto
Director of Global Programs at MIT Professional Education

"Solutions to clean energy must be innovative and collaborative. Data-driven analysis and improvements in technology are significant factors that will steer the future of our planet in the right direction. Generating energy that produces no greenhouse gas emissions from fossil fuels is a move in the right direction.."

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

1. **15% discount** for MIT Professional Education online courses and short-duration, in-person programs.
2. **Continued access to the virtual campus for six months after completion** of the course.
3. **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network.**
4. Invitation to the **MIT Professional Education Alumni LinkedIn group.**
5. **Networking opportunities** with other MIT Professional Education Alumni.

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PROFESSIONAL EDUCATION
Digital Plus+ Programs

Are you prepared to make data-informed decisions for clean energy solutions?

Frequently Asked Questions

global alumni
-excellence or nothing-

For more information, please contact us:

+1 617 870 0270

Email: **professionalprograms@mit.edu**

Website: **https://professionalprograms.mit.edu**